

An Experimental Online Judge System Based on Docker Container for Learning and Teaching Assistance

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Abstract—Programming Languages are core courses in computer science and computing-related disciplines, and there are increasing demands on improving the experiences, motivation and efficiency of programming language teaching and learning. In recent decades, online judge (OJ) systems have been popularly adopted in educational environments to support real-time learning feedback, and provide interactive programming practice. However, most existing OJ systems rely on significant resource for hardware virtualization, and suffer from long development cycles and extensive maintenances. This paper proposed a course-oriented OJ system based on the Docker Container techniques to significantly reduce the cost and maintenance of deployment of existing OJ system. The proposed approach is designed in the form of an experimental system, which is effective in stimulating the students to learn programming language and assist the instant assessment of coursework. Meanwhile, our proposed system can be integrated with major massive open online course (MOOC) systems for assessment. The efficiency and performance of our proposed system has been evaluated and tested by 1460 students in Nanyang Institute of Technology during term time between 2018 and 2019.

Keywords— Programming language learning; Online Judge; Docker Container; Experimental teaching system; MOOCs;

I. INTRODUCTION

The demand of improving teaching experience in Higher Education (HE) is increasing more than ever due to the advances in technology aided teaching. With the assists of technology, the teachers can reduce their burden of teaching and assessing the students' work [1]. Teaching involves various skillsets, such as public speaking and stress management, which can be improved through training and preparation [2]. But when it comes to students' assessment, it becomes time-consuming and low-efficiency due to repeatedly manual assessment and grading. The computer technology is driving the change of interactive teaching and automatic assessment, which increases the attention and focus of students, and engage them in learning process. Moreover, it enables students to get more involved in the learning process, especially outside of the class [4]. However, evaluating the assessments with full sincerity, and dedication is still an open challenge.

This paper studied the teaching in programming language, which is interdisciplinary at university level and students from multiple disciplines can study programming. Programming is one of the most sought-after skill and software engineers are in demand. The corresponding core courses are set in the universities in order to develop the

students' programming skills as their fundamental abilities, such as C++, Java, Python and R. Common objectives of these programming languages are 1) grasping the syntax of the language, 2) mastering the programming process and skills, and 3) applying the programming ideas to analyze and solve real-life problems. The most important purpose of programming language is problem-solving and practical ability. However, many students, especially as a beginner, find the process of learning a programming language a difficult and even unpleasant task [5, 6].

Many literatures have proposed the approaches of enhancing the experience and efficiency pertaining to the teaching and learning of programming language courses. One-to-one tutoring is the most efficient way. An individual's needs can be met by two-way dialogue between the teacher and the programmer. Moreover, live coding is of great help than using slides. It improves the student's response and they absorb more than the teacher's intention. Reference [7] argued that when learning programming, continuous practice is required to ensure that the knowledge retains and constructed a web-based programming-learning environment to support cognitive development. [8] showed practice testing and step-by-step practice can benefit learners of different ages and abilities, boost students' performanc. [9] designed a dialogue-based tool, Pro-Guide, to support students to create basic programs and motivate students to do many exercises and create many programs to improve their programming competences. [10] indicated the lack of a large amount of exercises is the key reason of failure in language programming courses. [11] mentioned that providing accurate and meaningful prompts based on individual students' test results has been recognized by researchers as being an effective approach. [12] proposed a suitable method that can increase the pedagogical effectiveness of existing instruction by using adaptive e-learning systems to provide individualized guidance and feedback to a student. [13] developed a CyberLiveApp with a secure sharing and migration Approach in order to enhance the efficiency of interaction. Apparently, it is more critical to attract the students to participate in the exercises and practice frequently.

Recent years, many universities use online course judge system to replace the traditional paper-based assessment approach to mark students' assignments. Major advantages of online evaluation over paper-based process are its increased efficiency and flexibility in online evaluation. Among various programming languages learning strategies, there is a type of assessment platform, called Online Judge (OJ), which is an online evaluation

system originally used to assess or score or grade the program in programming competitions, and can be used for practice [14]. The online evaluation system compiles the source code and executes it in the backend server.

In order to assist teaching, and improve students' learning experiences for programming language courses, this paper puts forward the idea of the combination of OJ's functionality and Docker Container technology, in the form of an experimental system. It aims to offer a potential problem-solving approach for computer science related major in HE.

The remainder of this paper is organized as follows: Section 2 summarizes existing OJ used for education purposes to improve the effectiveness of programming instructions. Section 3 describes our system architecture and function modules. Section 4 presents the implementation of our experimental system. Finally, Section 5 concludes the paper and future research directions.

II. RELATED WORKS

A. Assessments of programming languages courses

It is important to practice programming and find bugs through immediate feedback[15]. However, the traditional teaching methodologies and judging the students' submitted reports cannot meet the contemporary learning requirements of programming languages and has the drawbacks as follows:

- **The low efficiency of assessment:** Every student has their own coding style and habit of programming, there may exist different methods of problem-solving and implementation. Therefore, evaluation of the source code is a time-consuming and less-efficient task manually.
- **The lack of real-time evaluation results:** Unnecessary delay in the feedback of submitted reports or assignments result in confusion amongst programming. In the long run, it is not conducive to the cultivation of interest and enthusiasm in programming.
- **Existence of evaluation errors:** Loss of concentration, lack of understanding, misinterpretation may lead to evaluation error.
- **Less storage and analysis of evaluation results:** It is not convenient to keep the evaluation results of the paper and unfavorable to the subsequent data analysis. Evaluation is also a worthy indicator of student facing uphill task in programming.

B. Applications of OJ systems in Higher Education

OJ supports fully automated, real-time evaluation of source code, binaries or even textual output submitted by participants competing in a particular challenge [15]. It assesses the correctness of the submitted solutions based on predefined test sets and presents the online ranking of all participants in real-time during the ongoing contest [16]. As [17] concluded, OJ evaluation procedure consists of three steps: submission, assessment and scoring. According to its potential applications, there are six classes of OJ systems: (1) exclusive for contests (ACM ICPC-like), (2) educational purpose, (3) recruitment of employees, (4) evaluation of specialized algorithms, (5) crucial component of online compilers and (6) development platform. Correspondingly, any type of OJ systems

contains at least one of the following functionalities: online compilers, data mining, education, competitive programming, recruitment or development.

In this paper, we focus on the education-related or course-oriented OJ systems. The application of OJ systems in HE is at least since 1961 at Stanford University [16], then developed as a part of functions of Massive Open Online Courses (MOOC) to automatically judge the correctness of the exercises. Until 2016, there are at least 42,18,9,6,8,13 different kinds of OJ systems fall into the above-mentioned six catalogues respectively, in which there are 18 education-purposed OJ systems and 9 online compilers closely related to our discussions, partly shown in Table 1, 2 [17].

TABLE 1: EDUCATION-PURPOSED OJ SYSTEMS

Name	OSS	Problems	Compiler	Established	URL Address
Python Challenge	No	33	0	2005	http://www.pythonchallenge.com
Judge.org	No	2000	3	2006	https://judge.org
Embedded Security CTF	No	19	1	2014	https://microcorruption.com
Exercism	Yes	1450	3	2013	https://exercism.io
Codeboard	Yes	24000	3	2015	https://codeboard.io
Codechef	No	1500	3	2009	https://www.codechef.com
URI Online Judge	No	1170	2	2011	https://www.urionlinejudge.com.br
The AI Games	No	8	3	2013	http://theaigames.com
Code Fights	No	1250	3	2015	https://codesignal.com

Notes: (1) OJ system's name (2) Open Source Software(OSS) (3) The number of test cases (4) The number of supported programming languages, '0' does not support compiler, '1' supports a single specific programming language, '2' supports several popular programming languages, '3' supports a broad range of programming languages (5) The year of system's establishment (6) The system's URL address

TABLE 2: ONLINE COMPILES

Name	OSS	Compilers	Established	URL Address
Coding Ground	No	3	2006	http://www.tutorialspoint.com/codingground.htm
Codepad	No	3	2008	http://codepad.org/
Ideone	No	3	2009	https://ideone.com/
Online Compiler	No	2	2009	http://www.onlinecompiler.net/
CodeSkulptor	Yes	1	2012	https://www.codeskulptor.org/
Codeanywhere	No	3	2013	https://codeanywhere.com/
Codio	No	3	2013	https://codio.com/
C++ Shell	No	1	2014	http://cpp.sh/

As described in [15][16][17], the advantages of these educational OJ systems can be summarized as follows:

- Assess the submitted solutions automatically with higher accuracy and objectivity.
- Time required for evaluation is much shorter and save a large amount of scoring time.
- Timely feedbacks and almost instant answers stimulate students' enthusiasm in learning programming languages.
- Without the restriction of time and space, students can practice programming at anytime and anyplace via web browser.
- Without complicated programming Integrated Development Environment (IDE) installation or deployment, providing online code editor, and its compilers.
- Allow to define "classrooms" and trace the students' learning progress.
- Collect a large number of exercises classified by difficulty level or topic-oriented categories.

Undoubtedly, the OJ systems mentioned above have their own advantages on some aspects, i.e. accuracy of

evaluation, coverage of topics. Nevertheless, a few of them are course-oriented and lack features as follows:

- Systematic test instances which fully cover the syllabus of a specific programming language course.
- Effective anti-cheating mechanisms to prevent students from plagiarizing or pasting code.
- Visualization of statistical function according to every student's achievement.
- Implementation of the management of classes, courses, teachers, including the permission of different roles in the system

C. Docker Container technology in OJ

Virtualization is one of the core technologies of cloud computing. The server can be divided into multiple computers logically through virtualization technology, namely Virtual Machine (VM). Each VM can run a different operating system and applications independently, so it is possible to provide the different compilers and runtime environments of programming languages in VMs. As a lightweight virtualization technology, multiple instances of Docker Container can be run on a single host at the same time with customized resources. Then different compilers and runtime environments can be configured in each Container. Therefore, in the OJ or OJ-like systems, the user's source codes or solutions can be submitted to the corresponding Docker Container for compiling, running, evaluating in homogeneous environments. Additionally,, the results of evaluation could be returned to the client in real time. Lastly, container is destroyed and the resources are released.

As shown in Table 2, Codeanywhere is a cloud-based IDE where the users share a dedicated virtual development environment or a custom development environment created from virtual container. Coding Ground is also a cloud-based environment in which every program submitted by users is executed in a dedicated Docker-based container. C++ shell provides the functionality of compiling with a virtual sandbox created on demand. Online Compiler allows compiling the source code remotely [17]. From the analysis above, we can conclude the following advantages if we design the course-oriented experimental system with Docker Container technology:

- The underlying container module created by Linux sub-process and the adoption of pipeline for the data transmission can improve the efficiency and usability.
- Containerization provides reliable, secure compiling and runtime environment for each user with isolated, limited resources, i.e. CPU, memory, storage.
- Containerization can prevent the users from submitting abnormal code, modifying the testing environment, accessing restricted resources or occupying more resources while ensuring the program works properly.
- Docker container require no additional overhead with existing virtualization function of OS level isolation.
- The source code can be compiled, executed independently in the isolated containers which are created on demand.

In order to solve the existing challenges mentioned above, we proposed an experimental system that integrated the automatic evaluation and compiler together which is a course-oriented, OJ-like platform based on the Docker

container technology. This platform is tested and used by 1460 students at Nanyang Institute of Technology, China.

III. ARCHITECTURE OF THE EXPERIMENTAL SYSTEM

From technology's prospective, the framework of the experimental system includes three layers as shown in Fig.1. The upper layer provides user interface by browser, called Web Testing Terminal (WTT), which runs the services in Apache Tomcat Server and responses the users' request, i.e. compiling and running. The middle layer provides the virtual experimental with Docker Container which is the main part of the platform. The underlying layer provides data services, including Docker images ware, MySQL, Redis and Message Service.

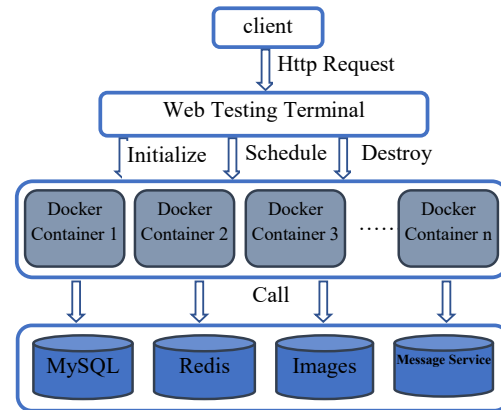


Fig. 1. Principle of the technology

Docker is an open source container engine project based on a lightweight virtualization technology and Apache2.0. Docker can rapidly and automatically deploy applications into the container, provide the isolation and security of container resources. Because the isolation is implemented by virtualization of OS layer, Docker container does not need the extra operating system overhead when it is running, so it can effectively improve the resource utilization. The Docker container structure is showed as Fig. 2.

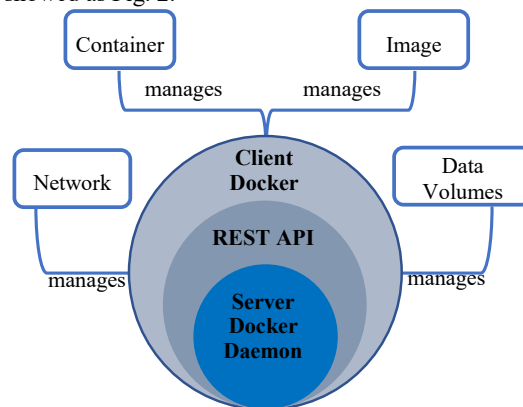


Fig. 2. Management Structure of Docker Container

Docker Daemon is responsible for listening the requests and performing appropriate operations in the backend of the experimental system. Docker CLI has the function of container management, i.e. creating containers from Docker image. The container can be a RFS (Root File System) which contains applications. Docker Daemon communicates with Docker Client through REST API to

create, destroy, reset the container. The Docker images are read-only templates which integrate Ubuntu OS environment. In this paper, the Docker image is built into the different compilation and execution environment, such as C/C++, Java, Python etc., then the corresponding Docker Container can be generated according to the type of source code submitted by the user. In addition, Docker Swarm can be utilized to balance, schedule and allocate the container resource pool. When the container server resource is not sufficient, it can dynamically add the server resource to expand the capacity of the resource pool.

From functionality point of view, the experimental system can be divided into three main components, which are shown as Fig.3.

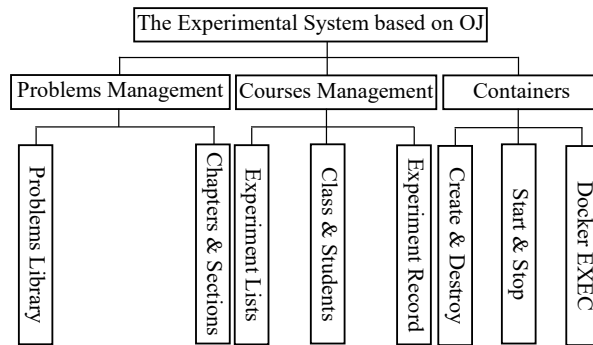


Fig. 3. Main modules of the experimental system

1) Problems management

The exercise problems attached to a specific programming language is one of the key functionalities of the system. Teaching team of the course can design appropriate exercise problems conveniently with this component, then an exercise problems library is developed, including description, code profile, score, keywords, input/output data. For example, in C++ or C, Code profile refers to a complete frame of Main Function Definition, in which the user needs to fill the relative code according to the requirement of the exercise problem. Keywords refer to a series of string which must be used in the users' source code in order to review their mastery of syntax or API. Additionally, this module can categorize the exercise problems according to the course's chapters and sections. Moreover, users with the permission can add, delete, query, and modify the exercise problems on demands.

2) Courses management

After creation of the exercise problems, the users who own the permissions can create different lists which includes certain exercise problems corresponding to a specific section, chapter, course, and create different classes, assign different lists to the classes on demands. They also can import students information to the system and attach to a specific class.

3) Containers management

This component entails the functions of creating or destroying container, starting or stopping the services. After login, the user initializes a fixed Docker container which save, compile, evaluate the submitted source code. Operating in exclusive Docker container have the feeling of running a standalone computer without mutual influence. In order to operate the Docker Container remotely by the user, Docker container server is pre-installed with Tester and tty.js. Tester is responsible for saving, compiling the

submitted source code, giving feedback timely. The script tty.js provides the Web Testing Terminal to the user. These two services are launched by the Docker EXEC.

In order to achieve server security, most OJ platforms prevent user code from accessing local resources directly, such as intercepting system calls from user programs using a "sandbox", so if the exercise problems have the operating of input/output, reading and writing files, network communication, the evaluation cannot be automatically performed. In this system an exclusive container can be assigned for each user to compile and execute the source code. If there is an illegal operation in the code, the container will be damaged, while the other users are not affected. It also solves the problems just mentioned above in other OJ systems. Moreover, Docker Container have the advantage of starting up in a few seconds, and the system can shut down idle containers to reduce resources consumptions.

B. Operation of the experimental system

As shown in Fig. 4, the implementation of the experimental system consists of front-end WTT, Database and Docker Container, in which Vue.js and Spring are responsible for the interaction with the users, all the related data is stored in MySQL.

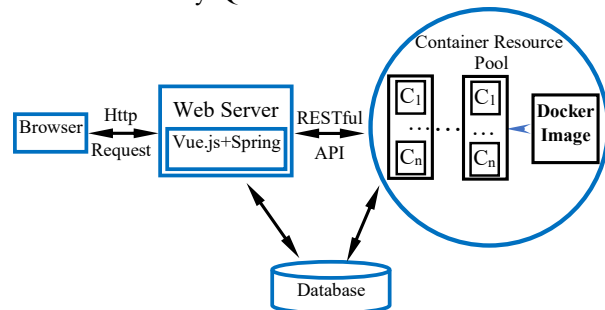


Fig. 4. Implementation of the experimental system

It is worth mentioning that when hundreds of users share a Docker server, the timely closure of the Docker container is essential. Normally, the Docker Container only runs for a few seconds to finish the corresponding tasks. If the source code submitted by the user contains an endless loop or an illegal input, the evaluation cannot be completed and the container will continue to run. In order to solve this problem, the system sets the maximum connection time between the web application and the container, once the threshold is exceeded, the exception handling code sends a message and closes the container. In addition, when the user remotely controls the container via tty.js, if the browser is not closed in time, the container will continue to run, when the user does not operate for a long time (30 minutes), web server will destroy the session, meanwhile, trigger the event to close the container.

C. Procedure of evaluation

As the critical part of the experimental system, the evaluation function is responsible for compiling, running and evaluating the submitted source code, the procedure is described as follows and shown as Fig.5:

1) After submitting the source code of the exercise problem via WTT, the source code and related information of the user is stored in the database, then the system generates the corresponding source file on the server according to the programming language used by the user.

2) User compiling requests are added to the end of the requesting queue, the main program traverses the request queue at fixed interval, takes out the request job, launches the container and loads the request job into the container.

3) According to the type of the source file, the container calls the corresponding compiler to compile the source file. If there are syntax errors, the compilation does not pass and the error information is returned in real-time. If the compilation passes, the executable file is generated.

4) After receiving the successful compilation message, the container calls the execution subroutine to run the generated executable file. During the process, the execution subroutine monitors the use of resources, including memory and CPU usage, if the threshold is exceeded, the operation is terminated and the corresponding information is output.

5) When the subroutine is finished normally, the output data is compared with the pre-defined input and output stored in database. If the result is inconsistent, the logic of the program is wrong, and the corresponding error information is returned, if the result of the comparison is consistent, the result corresponding to exercise problem is updated and the evaluation is passed.

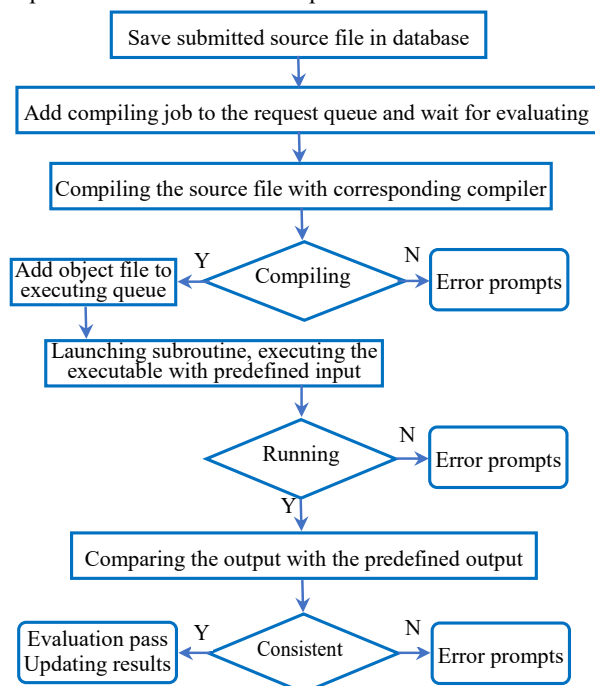


Fig. 5. The procedure of the evaluation

D. Key functions of the experimental system

According to the modules and analysis of the data flow, we described the key features of the experimental system as follows:

1) The users can edit, compile, run and evaluate the source code of the exercise via WTT. And WTT has the basic functions of IDE, i.e. syntax highlighting, code completion and compiling error prompts. Additionally, the user can login Linux environment via web terminal (tty.js) and observe the running process of the program. This online editing mode is not only convenient to use but also preventing plagiarism (disabled the pasting code function in the browser), i.e. copying and pasting the source code. The web editor is shown in Fig. 6.

2) The exercise problems of this system can cover more knowledge points of the course. For the sake of the security of the server, OJ platform usually prevents the submitted source code from calling API, such as network communication. As results of assigning a Docker container for each user to compile and execute the submitted source code, then the above restrictions can be resolved.

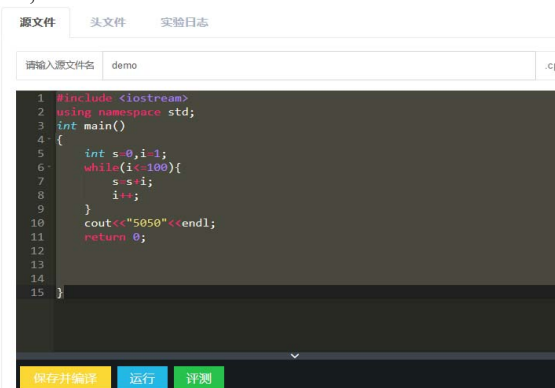


Fig. 6. Web Testing Terminal of the experimental system

In addition, the system also supports the evaluation of multiple source files, that is to say, a complex exercise problem can be divided into multiple source files, then compiled and evaluated one by one.

3) The system can record the students' experimental process in the log table, including the time of submitting the source code each time, the evaluation results, and the error information. These data can track the process of each student's learning growth and lay the foundation for the subsequent data analysis, and targeted guidance. The test graph is shown as Fig. 7.

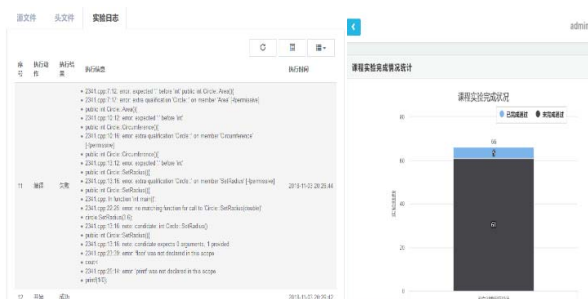


Fig. 7. The log information of experimental process



Fig. 8. The implementation of curriculum management

4) The system can evaluate the submitted source code and generate the exercise report automatically. The system can evaluate the source code through keywords matching or 'black box test', save the score of each exercise problems. Finally, the teacher can export the

transcript of all exercise problems by clicking a button easily. In this way, teachers can put more attention on their instruction or guiding students.

5) The system provides the role management module, including the program leader, the teacher, and the student. The permission of roles is depicted in Table 3, and the implementation of curriculum management is shown as Fig.8.

TABLE 3: THE PERMISSION OF ROLES

Roles	Permissions and Description
Program Leader	Course Management: CRUD information of course, such as ID, name, chapters.
	Teacher Management: CRUD information of teacher, assigning the courses and classes for teachers
	Student management: importing information of students by fixed template
Teachers	view the teacher's curriculum and corresponding classes
	schedule the experimental deadline
	CRUD experimental information, such as title, requirements, I/O test sets. Class management, i.e. score exporting, progress checking, reset container.
Student	Login the system with ID and password, select experiment attached to specific course, chapter and sections, editing, compiling, operation and evaluation of the exercise problem

IV. CONCLUSION AND FUTURE WORK

OJ systems provide a platform for testing learners' mastery of programming ability and can automatically evaluate the solutions submitted by users in real-time feedback with the advantages of high speed and accuracy. Docker container is a lightweight virtualization technology, which has the advantages of high security, high startup speed, faster development and deployment cycles. This paper proposed a course-oriented experimental system by combining the advantages of OJ systems and the Docker container to evaluate online submissions of programming courses. Our proposed system allows students to login anytime through the proposed experimental system and its implementation. With specific course requirements and exercise problems, edit, save, debug, run, evaluate the source code of the program and get timely feedbacks during debugging. With our proposed system, students' interest and enthusiasm in programming learning can be motivated. It can also greatly reduce the time of guidance, review and feedback of students' programming training, devote more time and energy to organizing teaching contents efficiently, design program-training topics, and effectively improve the teaching quality. As a part of our future directions, we plan to:

1) Design more effective programming training topics, namely exercise problems, for other courses in HE.

2) Optimize the program evaluation algorithm to support the comprehensive evaluation of the source code in terms of the syntax, logic, input and output.

3) Visualize the growth curve of the students' understanding degree of curriculum knowledge, and their programming ability.

ACKNOWLEDGMENT

This work was conducted in collaboration with Data Science Research Centre at University of Derby

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